

UNIVERSAL PLASTIC SHREDDER V2.0

Electrical Control & Power System

Bao Nguyen • Fearghus Tyler • Yaqoub Rabiah • Fox Kang

Portland State University • MCECS • Department of Electrical & Computer Engineering

Sponsor: MME / Ichor Systems (Bridget Lannigan) | Advisors: Prof. Andrew Greenberg, Dr. Jonathan Bird (EE), Robert Paxton (ME)



Maseeh College of Engineering
and Computer Science

PORTLAND STATE UNIVERSITY

The Problem

Ichor Systems generates large volumes of thick rigid plastic (PVC, PP, LDPE) during semiconductor equipment manufacturing. Off-the-shelf shredders are either too small for the material (0.25–0.5 in. sheets, 4–12 in. wide) or priced far above the sponsor's per-line budget. The EE team designed and built the electrical control and power system for an industrial-grade plastic shredder; the mechanical structure is delivered by a separate ME team.

Requirements

Must:

- Shred plastic to ~1 in. pieces.
- Drive a 3–5 HP motor at 70–90 RPM through a controllable VFD.
- Follow NFPA 70 (NEC); include E-stop, overload, and one safety interlock.
- Provide an HMI for operator control and status.
- Stay within the \$1,000 EE-team budget.

Should: status / fault lights, NEMA enclosure, labeled controls.

May: read-only wireless monitoring, data logging, capacitive-touch safety.

System Architecture

The control system is built end-to-end on AutomationDirect industrial hardware – no hobby-grade microcontrollers are in the safety or control path. A PowerTec 71008 disconnect feeds a Mean Well NDR-480-24 24 VDC supply, a Mitsubishi SD-N35 contactor (E-stop hard-wired in series with its coil), and a Huanyang HY02D211B-T VFD. The VFD drives a 2 HP IronHorse MTCP2-002 premium-efficiency TEFC 3-phase induction motor (230 V, 5.93 A FLA, 1735 RPM, 145TC frame, VFD-rated 1.0 SF) over U/V/W via blue, white, and red 4 mm banana sockets.



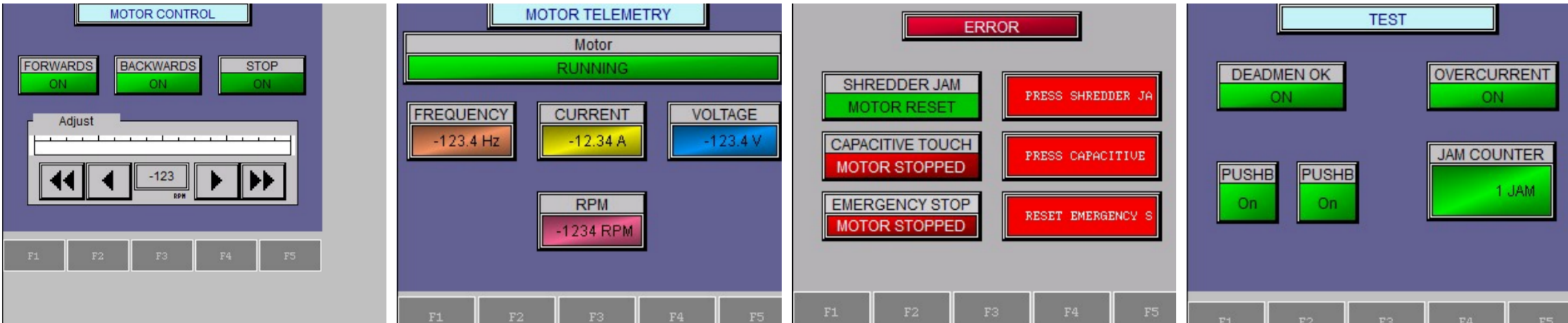
Operator face (left), panel interior (middle), and system block diagram (right): HMI, deadman / fwd / rev / E-stop, DirectLOGIC 205, Huanyang VFD, 24 V supply, SD-N35 contactor, terminal blocks; block diagram shows the signal path from AC in through the safety chain to the VFD and motor.

PLC Control

The PLC is the system controller. A DirectLOGIC 205 D2-09B-1 rack holds the H2-DM1E CPU and five I/O modules: D2-08ND3 (8 DC inputs), F2-08AD-1 (analog current input), F2-04RTD (RTD), D2-08TR (8 relay outputs), and F2-08DA-2 (0–10 V analog output to the VFD). The 31-rung Do-more program implements a deterministic state machine plus a 3-strike jam recovery sequence and a hard-lockout path.

HMI

AutomationDirect EA1-T4CL C-more Micro-Graphic 4 in. touch panel running HMI_413.mgp, serial to the H2-DM1E CPU. Four screens hold 27 PLC tags.



Motor Control • Motor Telemetry • Error / Alarm • Test.

Capacitive-Touch Safety (CTSI) Module

A contact-free operator-presence sensor designed and built in-house by the Shredder EE team. An ESP32 reads a capacitive electrode; on a touch event the ESP32 drives an LED through an optocoupler that closes the 24 V PLC input x7. Power is taken from the panel 24 VDC rail through an LM2596 buck converter to the ESP32's 3.3 V rail.

[CTSI module photo]

Drop Photo.CTSI.Module.jpg into
src/Pictures/ to insert it here.

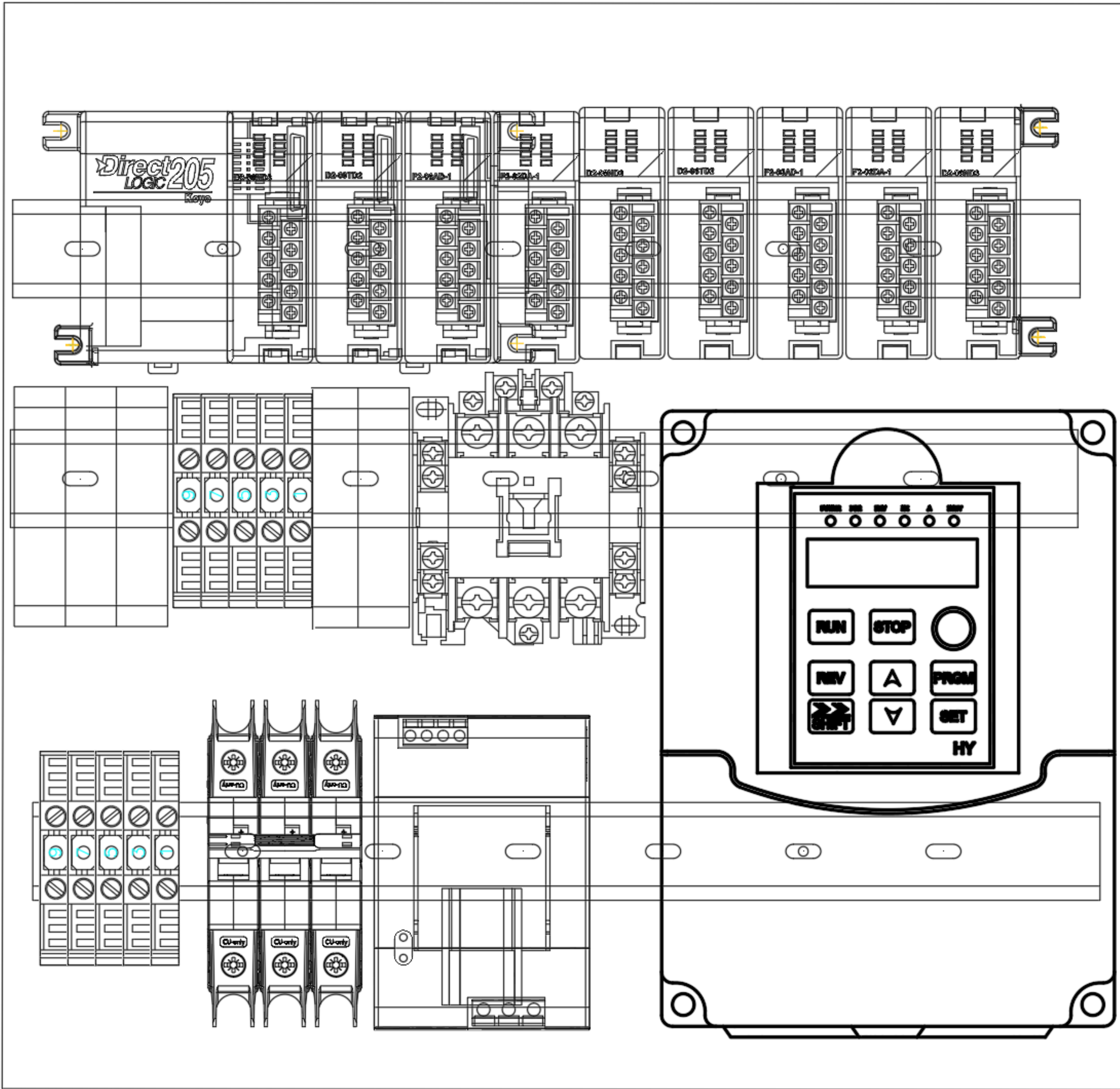
ESP32 + LM2596 buck + optocoupler module wired to PLC X7.

Ladder response (rung 28): touching the electrode latches Y4 (fault LED), C15 (STOP.MOTOR), C21 (CTSI.TRIG), and C22 (CHANGE_SCREEN) – motor stops, fault LED on, HMI jumps to ERROR, all independent of the deadman / E-stop chain.

Status: bench-validated against the PLC; full machine-level verification on the enclosed prototype pending.

Panel CAD Layout

The panel layout was drawn in AutoCAD Electrical (electrical/*.dwg, *.aepx) before assembly. Components are arranged on the back-plate as built: DIN-rail terminal blocks along the top, the DirectLOGIC 205 PLC rack mid-left with modules D2-08ND3 / F2-08AD-1 / F2-04RTD / D2-08TR / F2-08DA-2, the Mitsubishi SD-N35 contactor with the fused 24 V branch beside it, the Huanyang HY02D211B-T VFD at the right, and the EA1-T4CL HMI cut-out on the door.



AutoCAD Electrical panel back-plate: terminal blocks, PLC rack + modules, SD-N35 contactor, branch fusing, VFD, and HMI cut-out.

Safety & Compliance

- E-stop:** Category 0 stop. Eaton E22B1 NC mushroom in series with the SD-N35 contactor coil. Drops the contactor independently of the PLC.
- Two-hand / deadman:** both pushbuttons must be held to enable the motor (PLC X2 AND X3).
- Overcurrent / jam:** VFD over-torque trip (PD124 = 150%, PD125 = 3 s) on PLC X5; ladder runs the 3-strike unjam routine.
- Enclosure:** Vorksmen SEB001 steel, NEMA Type 12.
- Compliance:** NFPA 70/79, NEMA 250 / ICS / MG 1, ISO 12100, 13849-1 (Cat B-1 / PL b-c), 13850, 13851, OSHA 1910.147.

Results

All EE “Must” requirements met for the items the EE team controls: VFD motor control, NFPA-70 design, E-stop + overload + safety interlock, HMI control & status, and the \$1,000 budget.

Constraint: the ME team did not complete the mechanical shredder build by end of term, so full machine-level integration testing (loaded shredding of plastic samples on the actual frame) was not possible. All EE validation is bench-only against the VFD and motor stand-in; the panel is ready to integrate as soon as the mechanical prototype is delivered.

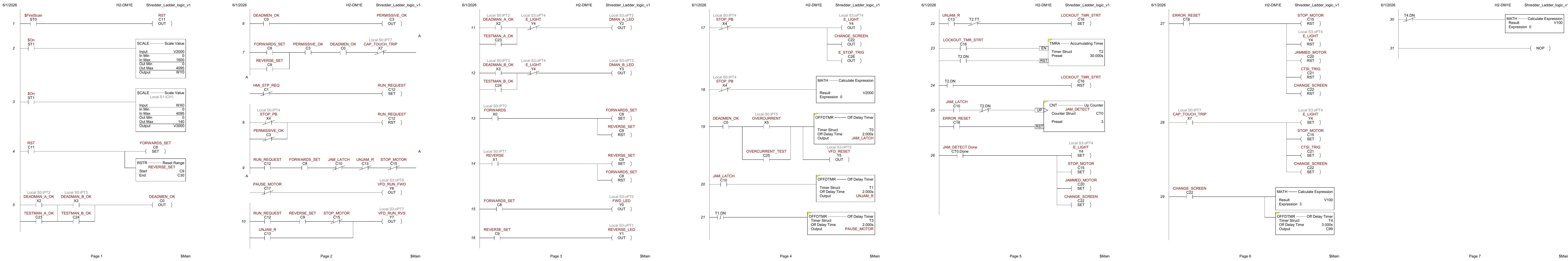
Working:

- Panel built, wired, bench-validated. All 31 rungs program and scan correctly.
- All four HMI screens operate; tag database matches the PLC.
- Speed reference verified: HMI V2000 → SCALE → F2-08DA-2 CH1 → VFD VI.
- Jam-recovery verified: overcurrent on X5 → 2 s reverse (T1) → coast-to-stop buffer (T3).
- Three-revision point list signed off by Fearghus Tyler.

Wired and ready, programming pending: Everything below is fully wired into the panel and the ladder program; what's left is configuration / firmware work, not hardware.

- CTSI module not yet integrated** – the capacitive-touch breakout and its microcontroller were destroyed during bench bring-up. The X7 input, optocoupler interface, and rung-28 ladder response are still in place and ready for the replacement module to drop in.
- Jam-count telemetry not finished** – counter C10 / N0 increments correctly inside the PLC but is not yet surfaced on the HMI for live readout.
- Voltage / current telemetry not working** – plan was to take the VFD's frequency and motor-current analog outputs into the PLC and back-calculate the derived voltage, RPM, and status fields shown on Screen 2. The path VFD VO → F2-08AD-1 CH1 → WX0 → V3000 is wired and rung 3 scales it, but VFD analog-output parameters and the PLC scaling still need to be revisited before the live values stabilize.

Complete Ladder Logic — 31 rungs across 7 pages



Pages 1–7: power-up init / deadman • RUN.REQUEST gating / Y6 Y7 outputs • indicator LEDs / direction latches • E-stop / jam detection / unjam stroke • lockout timer / jam counter • fault reset / cap-touch / HMI screen change • screen reset / end of program.

Project By the Numbers

31 ladder rungs in production.
27 HMI tags across 4 screens.
14 component sheets in the point list.
3 strikes before the motor locks out.
2 s unjam reverse stroke.
150% over-torque threshold on the VFD.
\$969.88 spent against the \$1,000 control-panel budget (\$30.12 under).
Lots of “no, lowballer – I know what I got” from vendors during procurement.

Hardware Stack

PLC: DirectLOGIC 205 + H2-DM1E.
HMI: C-more EA1-T4CL.
VFD: Huanyang HY02D211B-T.
Motor: IronHorse MTC2P-002 (2 HP, 1735 RPM, 145TC, TEFC, VFD-rated).
PSU: Mean Well NDR-480-24.
Contactor: Mitsubishi SD-N35.

Future Work

Production motor selection (3–5 HP) supersedes PD141–PD144.
Mechanical integration and full shred-load run pending – ME team did not deliver the prototype by end of term.
Capacitive-touch (CTSI) module on x7: machine-level verification pending.
Modbus link to the VFD for richer telemetry (deferred – the Modbus module exceeded the \$1,000 control-panel budget, so the analog VO / VI path was used instead).

Acknowledgments

Thanks to **Bridget Lannigan** (MME / Ichor Systems) for sponsoring the project; to advisors **Prof. Andrew Greenberg**, **Dr. Jonathan Bird**, and **Robert Paxton**; to the **ME capstone team** for the mechanical design and ongoing integration coordination; and to the **PSU Power Engineering Laboratory** for bench access.

ECE 412/413 Capstone Poster Session • PSU MCECS • June 12, 2026